Artificial urinary sphincter surgery

Cue cards: what to say on each slide

Prepared for Mr Andrew Baird, Mr Abu Yousif, Nity G | 20 October 2026

One card per slide: the spoken lines in order, each one a sentence you can say as written, ending with the line that carries the room to the next slide. Pauses are in square brackets. The clock under each title is the running time at which the slide should start; it is not on the slides. The full narration, the examples and what each study does on its slide are in the presenter script; this is the short version for the lectern.

Decide before theatre

00:00 – 14:00 · 14 minutes · slides 01 to 09

Section slide: Our first question for every man: can he get useful continence while storing and emptying safely, and work the device?

01 The decision before theatre

00:00 – 01:30 · 1:30

- An AUS can be a perfect implant and a disappointing treatment: the cuff coapts, the pump cycles, and he still stays at home.
- Today we follow the decisions that separate those outcomes: what he wants, then the bladder, then the urethra.
- One question keeps coming back, in four words: implant now, optimise first, investigate further, or reconsider.
- We will put it to four fictional patients, and hand the decision to you each time.
- Next: before we lean on any evidence, let us be honest about what kind of evidence it is.

02 Evidence and device boundaries

01:30 – 03:00 · 1:30

- Thank you to Rigicon for organising the meeting; the clinical discussion stays independent of the sponsor.
- Guidelines give us a pathway. Expert agreement is not a comparative trial.
- A manufacturer's instructions tell us how a product is meant to be used, not whether it beats another.
- BAUS sets out the UK pathway; we compare it with EAU and AUA where they differ. ICS and Asia-Pacific are AMS 800 documents.
- Where the evidence stops, we will say so. Next: start where the patient starts, with the leak.

03 Leakage mechanism and patient goals

03:00 – 04:45 · 1:45

- Start with a bad day. Standing, walking, coughing? Does an urge come first? Dry at night?

- "Mixed" is a description, not an explanation. Demonstrate stress leakage with a filled bladder.
- Pad count tracks leakage well overall, but two to four pads a day hides a wide range of weights. Weigh when unsure.
- Then ask what improvement would be worth an implant: the golf course, work, one precautionary pad.
- Next: once we know what he wants back, the question is when, and whether an AUS is the only route.

04 Timing after prostate treatment

04:45 – 06:15 · 1:30

- Do not let conservative management drift, and do not interrupt a convincing recovery.
- BAUS: at least six months for severe leakage, twelve for mild to moderate. AUA: from six months if not improving, offer at one year.
- A sling remains an option for milder leakage and good tissue. Radiation and severe leakage move us towards AUS.
- MASTER at 24 months: the sling stayed non-inferior, but further surgery was eleven per cent after sling, two after AUS.
- Next: timing assumes he can use the device. We test that; we do not ask about it.

05 Pump use over time

06:15 – 07:30 · 1:15

- Put a demonstration pump in his hand and watch. Arthritis, tremor and weak pinch only show when he tries.
- Access matters as much as strength: can he reach the agreed side of the scrotum while seated?
- Check he understands cycling and deactivation, and what to tell the next clinician before a catheter.
- Frailty is more than age: in the Medicare cohort it raised early complications and later revision, independently of age.
- Next: with the patient assessed, two investigations change the plan most often. First the cystoscope.

06 Flexible cystoscopy before implantation

07:30 – 09:15 · 1:45

- AUA recommends cystourethroscopy before SUI surgery; BAUS selects it within its urodynamics pathway; EAU is selective.
- Our preference is to look in advance, so a surprise is discussed before the day of implantation.
- We are looking for stricture, anastomotic stenosis, bladder-neck contracture, foreign material, stones, tumour, previous erosion.
- With a narrowing, add flow, residual, site and length. Do not turn a diagnostic scope into an unplanned dilation.
- Next: the second investigation is where the guidelines genuinely disagree, so we name the positions.

07 Urodynamics: BAUS, EAU and AUA

09:15 – 11:00 · 1:45

- BAUS: video-urodynamics for surgical candidates. EAU: urodynamics before invasive treatment, weak recommendation. AUA: when it changes diagnosis or counselling.
- Yield is highest when stress leakage is uncertain, urgency dominates, emptying is poor or compliance worries us.
- Before requesting the test, state the question it should answer and what would change.
- No trial shows routine testing improves AUS outcomes for everyone. The 2026 AUA abstract revisits this; results not yet available.
- Next: whichever pathway you follow, the study you get back will be imperfect. Here is how we read it.

08 Interpreting an imperfect study

11:00 – 12:45 · 1:45

- A man who leaks throughout filling never reaches a real storage challenge: a small capacity may be leakage, not a stiff bladder.
- Look at the traces, subtraction, filling rate and why filling stopped. A compliance number printed during artefact has no authority.
- Weissbart: 32 of 169 men did not leak with the catheter in, and every one leaked once it came out.

- Five steps: symptoms and demonstrated SUI; urine, diary, flow, residual; urethral anatomy; the chosen UDS strategy; then classify storage, emptying, outlet.
- Next: when the assessment is complete, we consent, and consent is where the numbers need their denominators.

09 Consent: improvement and lifetime burden

12:45 – 14:00 · 1:15

- Consent covers residual leakage, urgency, erosion, infection, mechanical failure and revision.
- AUSCO, AMS 800: 61 of 101 men used zero pads at twelve months. Single-arm, manufacturer-funded, missing data.
- ContiClassic: 93.2 per cent device survival at twelve months in 116 recipients. Device survival, not continence, no comparator.
- The regret study: reoperation was the only independent predictor, and benefit faded with time. Future surgery belongs in consent from the start.
- Next: that is the decision before theatre. Our first fictional patient puts it to work, and the decision is yours.

The bladder

14:00 – 22:00 · 8 minutes · slides 10 to 14

Section slide: Three bladder problems, three different decisions: overactivity, weak contraction, poor compliance.

10 Case 1: six pads and urgency

14:00 – 15:45 · 1:45

- Six pads on an active day, clear stress leakage, urgency leakage with detrusor overactivity, safe pressures, good emptying, patent urethra.
- Implant now or treat the bladder first? [Pause eight seconds.]
- Overactive bladder is a symptom syndrome; DO is an observation. The systematic review found no grounds to refuse surgery for DO alone.
- It does not promise urgency resolves; in one reintervention cohort, DO went with earlier reintervention.
- Our approach: treat the storage symptoms, agree urgency as a separate goal, reassess. For ContiClassic the IFU requires the bladder condition controlled or resolved.
- Next: so how do we treat that urgency without creating an emptying problem?

11 Urgency treatment and emptying

15:45 – 17:15 · 1:30

- Reversible contributors first: fluids, caffeine, constipation, infection, medications. Then bladder training and a drug chosen for his cognition and residual.
- Botulinum toxin works for refractory urgency, but could he catheterise, and how would repeat injections pass a future cuff?
- Lai: most men with mixed symptoms still had urgency after a successful AUS. Improvement in one is not failure in the other, if discussed.
- Severe urgency after surgery is reassessed: infection, residual, device function, before escalating drugs.
- Next: our second patient has the opposite problem. He cannot empty well even before a cuff.

12 Case 2: weak contraction and residual urine

17:15 – 19:00 · 1:45

- Stress leakage, but he voids slowly, strains and leaves urine behind; pressure-flow shows a weak contraction.
- Is that enough to refuse him an AUS? [Pause eight seconds.]
- First exclude a narrowing, review drugs and neurology, and compare the laboratory void with his real void.
- The DU cohort looked reasonable, but it excluded early revisions and cannot be stretched to severe retention.
- Our answer: investigate and build a reliable emptying plan. If it holds, implant may be reasonable; if he cannot empty or manage the rescue, reconsider.
- Next: if his rescue plan is a catheter, we must be clear what catheterising through a cuff involves.

13 Catheterisation with a cuff in place

19:00 – 20:30 · 1:30

- Intermittent catheterisation does not exclude an AUS, but it changes the consent and the operation.
- A small series supports feasibility in selected men; it does not prove equivalent safety.
- The rule: cuff fully deflated, device deactivated, smallest atraumatic instrument, stop at unexplained resistance.
- A deflated cuff's internal diameter is not a catheter-size chart. The 2026 survey found real confidence gaps on exactly this step.
- Prolonged urethral catheterisation is linked with erosion, so repeated or long drainage needs a specialist plan.
- Next: one bladder problem can look like success. A dry patient can still be unsafe.

14 Detrusor overactivity and poor compliance

20:30 – 22:00 · 1:30

- Poor compliance is the pressure-volume relationship during filling, not just a contraction. Left: transient contraction. Right: sustained rise.

- Check the trace: sustained rise, adequate filling, valid subtraction. Think radiation, fibrosis, neurology, chronic leakage.
- Assess renal function and the upper tracts when storage pressure worries us; optimise and repeat before raising outlet resistance.
- Lai 2009 made the point: pad results can look good while poor compliance still threatens the kidneys.
- Our four decisions are now clear: implant, optimise, investigate, reconsider. A framework, not a validated score.
- Next: that closes the bladder. The urethra is next, and the first job is to name the narrowing.

The urethra

22:00 – 29:00 · 7 minutes · slides 15 to 19

Section slide: Name the narrowing, prove the outlet is stable, and plan the tissue the cuff will need.

15 Three different sites of narrowing

22:00 – 23:15 · 1:15

- "Stricture" hides three different operations: anterior stricture in the spongiosum, bladder-neck contracture with prostate present, VUAS after prostatectomy.
- Review every dilation, urethrotomy and reconstruction, with the operative reports. How fast did it recur? Does patency depend on calibration?
- Stable means a credible interval without recurrence, with symptoms, flow, residual and endoscopy agreeing. A scope that passed once is not that.
- Think ahead: a man likely to need more incisions or surveillance needs an access plan before a cuff complicates it.
- Next: stability is a finding, not a date. Here is what the three guidelines say about the interval.

16 Reassessment and urethral stability

23:15 – 24:45 · 1:30

- BAUS: at least six months without recurrence after treating a post-prostatectomy narrowing. Consensus, not a trial.
- AUA: interval cystoscopy at least four to six weeks after treating BNC or VUAS, a reassessment point, not a clearance.
- EAU: three to six months after redo anastomosis belongs to that reconstruction only.
- Staged treatment lets us watch for recurrence and spares a new cuff from instrumentation. Synchronous treatment is for selected expert series.
- A man who restenoses every few weeks is not low risk because the calendar reached a date.
- Next: our third patient brings all of this together, with radiation added.

17 Case 3: radiation and recurrent VUAS

24:45 – 26:15 · 1:30

- Prostatectomy, salvage radiotherapy, repeated treatment of an anastomotic stenosis. He leaks when it is open; it keeps closing. He wants to be dry now.
- Would you put an AUS into that? [Pause six seconds.]
- Our answer: defer and get a reconstructive plan. Patency is not durable and more endoscopy is likely; check compliance and radiation cystitis.
- Radiation roughly doubles the odds of erosion in pooled data. That is counselling, not his personal probability. The 2025 timing study sets no waiting rule.
- Reconstruction, demonstrated stability, then AUS may be reasonable. In a devastated outlet, diversion can be the better life.
- Next: suppose he gets his reconstruction. What does a repaired urethra mean for the later operation?

18 AUS after urethral reconstruction

26:15 – 27:45 · 1:30

- Previous urethroplasty is several operations: transected or not, graft, posterior repair, prior erosion, fistula. Where is the viable spongiosum now?
- Preserve what remains, do not extend mobilisation through a repair, and stage if a safe plane cannot be found.
- Davis: explantation-level complications 56 per cent after transecting versus 24 per cent after non-transecting repair. The 2026 meta-analysis points the same way.
- Observational and confounded; it supports a detailed history and careful counselling, not blame.
- Plan the reconstruction and the continence surgery together. Next: when the dorsal plane is hostile, one option goes through the corpora. It has a price.

19 Transcorporal placement: role and costs

27:45 – 29:00 · 1:15

- Transcorporal placement avoids a dangerous dorsal plane and puts tunica under part of the cuff. Plausible, not proven protection.
- The data point both ways: Redmond favoured it numerically; Kurtzman found more explants; the 2025 review pooled less erosion but more revision.
- Surgeons choose it for the worst urethras, so confounding runs in both directions.

- Discuss corporal bleeding, erectile consequences and any future penile prosthesis. Size by the device protocol, with no improvised correction.
- If the proposal is just to squeeze a failing urethra harder, look again at the diagnosis and the endpoint.
- Next: now a different kind of decision, the things we do before the skin is opened.

Preparation without ritual

29:00 – 36:00 · 7 minutes · slides 20 to 24

Section slide: Before we debate how long anything takes, we identify the activity, the product and the evidence.

20 The fifteen-minute wash question

29:00 – 30:15 · 1:15

- Is a fifteen-minute wash necessary before an AUS? We found no direct evidence for a universal requirement.
- First name the activity: patient washing, skin preparation, hand antisepsis, irrigation and device preparation. Each has its own instruction.
- BAUS recommends a five-minute topical field scrub by Delphi, openly extrapolated from other implant surgery. Not fifteen minutes.
- Fifteen minutes cannot rescue missed skin, the wrong product, pooled alcohol or a contaminated field. Correct and reproducible beats long.
- Next: what the studies can actually support is narrower than the folklore, and here is the best of them.

21 Skin culture reduction and clinical infection

30:15 – 32:00 · 1:45

- Yeung: 100 initial genitourinary prosthetic cases; positive skin cultures 8 per cent with chlorhexidine-alcohol versus 32 with povidone-iodine.
- A microbiological surrogate in a mixed population. It did not show fewer AUS infections and tested no duration.
- The later povidone comparison showed no advantage for an intensified protocol, and never compared fifteen minutes with shorter.
- Use an alcohol-based preparation where the product allows it; check allergy, permitted sites and complete drying before diathermy.
- Do not remove hair routinely; clip on the day if you must. Next: skin is one disagreement. Urine and antibiotics are another.

22 Urine and antibiotic disagreements

32:00 – 33:30 · 1:30

- Treat symptomatic infection and reassess. Asymptomatic bacteriuria is different.

- BAUS asks for a urine sample and treatment. IDSA suggests against screening or treating for an AUS alone, on very low-quality evidence.
- Agree the pathway with microbiology; choose prophylaxis by policy, isolates, allergy and renal function; complete it before incision and redose.
- Adamsky: explantation 3.9 versus 4.0 per cent with and without postoperative oral antibiotics. The ContiClassic IFU describes a longer course.
- We verify current labelling and write a local protocol. Next: once the patient is prepared, the remaining risk is our hands and the implant.

23 Contamination control during implantation

33:30 – 35:00 · 1:30

- Prepare before opening the implant, protect a handling area, minimise skin contact, cut traffic and hand-offs.
- Fresh gloves before the device and after any concern: a practical control, not a proven step. Minimal-touch series compare eras.
- Separate wound irrigation from device dipping and filling. NICE does not support irrigation solely to prevent SSI.
- The gentamicin-withdrawal cohort: one infection in 59 with, two in 114 without. Three events prove nothing either way.
- For ContiClassic, follow the verified instructions and obtain the operative protocol; no remembered AMS 800 recipe, no improvised bath.
- Next: all of this collapses into one checklist the whole team runs before the implant is opened.

24 The pre-implant theatre checklist

35:00 – 36:00 · 1:00

- Before opening: patient, operation, tissue concerns, stop conditions; infection review, prophylaxis given, redosing plan; preparation suitable and dry.
- Then the exact device and components, cuff sizes, pressure range, current instructions, who prepares it, and the drainage plan.
- Decide now what happens if the urethra is injured or sterility is lost. A checklist makes that response predictable.
- The last row matters: a fixed fifteen minutes, the best irrigant and prolonged prophylaxis sit there as unresolved, not required.

- Next: into theatre. The next eight slides follow the operation in order.

The operation

36:00 – 50:00 · 14 minutes · slides 25 to 32

Section slide: Preserve tissue, measure what the cuff will enclose, follow the real device protocol, verify before closing.

25 Positioning and exposure

36:00 – 37:15 · 1:15

- Expose the bulbar urethra while protecting pressure points and respecting hip mobility; check before anaesthesia if movement is limited.
- A single-cuff perineal approach has guideline support for the standard bulbar implant.
- Plan balloon and pump sites now: pelvic surgery, mesh, scars and an existing prosthesis change the plan. Have the records.
- If exposure is poor, improve the incision, retraction or position. Traction to compensate for a small space goes straight to the urethra.
- Next: with the urethra exposed, which tissue will the cuff sit on, and how much of it do we disturb?

26 Bulbar dissection and tissue preservation

37:15 – 38:45 · 1:30

- The goal is a cuff around a viable segment, not a skeletonised urethra. The proximal bulb often offers good spongiosal support.
- The dorsal plane against the corpora is where a small error becomes a full-thickness injury. Dissect under direct vision; never pass an instrument blind.
- Preserve spongiosum and attachments, mobilise only what the cuff needs, control bleeding without indiscriminate diathermy.
- If the plane disappears, stop: improve the view, change the site, or get reconstructive help. Transcorporal is a considered plan, not a rescue.
- Next: once the segment is chosen, we measure it, and measurement is where the most avoidable errors live.

27 Cuff measurement and fit

38:45 – 40:30 · 1:45

- Measurement fails when the tissue measured is not the tissue the cuff will enclose: traction stretches, a tight tape compresses, fat and scar inflate.
- Release the traction, confirm the plane, use the manufacturer's sizing method, recheck anything implausible.
- Too small: compression, retention, injury. Too large: poor coaptation, leakage. Neither is fixed by a stronger balloon.
- Four of the eight revisions in the first ContiClassic series were sizing errors. That is why we recheck, not which strategy is best.
- Next: sometimes the dissection or the measurement reveals something that changes what today's operation is for.

28 Urethral injury or contamination

40:30 – 42:00 · 1:30

- Suspect injury when the plane is wrong, the wall looks breached, fluid escapes or instrumentation was difficult. Test and scope; a negative test does not license force.
- Recognised injury changes the objective: stop implantation, define it, repair, drain, explain, reassess healing before any reimplant.
- Contamination: stop handling, identify what is contaminated, replace disposables, restore a clean setup, or stage.
- These are single-use implants. A dropped cuff does not come back after an antiseptic soak. Finishing without an implant can be the right outcome.
- Next: if the cuff is safely placed, two separate balloon decisions remain: where it goes, and which pressure.

29 Balloon pocket and pressure selection

42:00 – 43:30 · 1:30

- A routine retropubic pocket becomes hazardous after pelvic surgery, radiation, mesh or previous implants. Never overcome resistance with force.
- An ectopic or submuscular pocket can be right in experienced hands, with its own risks: malposition, palpability, herniation.
- Pressure is a separate decision. ContiClassic labels sixty to sixty-nine; AMS 800 consensus commonly sixty-one to seventy. Not interchangeable.
- The public IFU confirms saline filling but sends detail to a separate OR protocol; confirm volumes and selection before surgery.

- Pressure never substitutes for a poor cuff fit. Next: the third component is the one the patient touches every day.

30 Pump position and patient access

43:30 – 44:45 · 1:15

- The pump position is part of the result. He has already shown us where he can reach and which hand he uses.
- Dependent and accessible, but the pocket must fit: too much dissection invites migration; too little causes pain and pressure.
- Keep it distinct from the testis and away from thin skin. Check how it sits standing and whether tubing pulls it upward.
- A pump that is easy on a table is not proof it works in a swollen scrotum. Check again at activation, with him doing it.
- Next: then we finish the system, and we hand over the risk with it.

31 System checks and handover

44:45 – 46:00 · 1:15

- Trace every tube: no tension, loops, kinks or entrapment. Keep needles away from the components.
- Prepare and connect by the device's own protocol, cycle it, then confirm the cuff is open and the device deactivated. Pressing a button is not enough.
- A small atraumatic catheter, placed gently, removed early after an uncomplicated implant. No improvised size.
- Record model, cuff site and size, balloon spec and position, deactivated state, catheter plan and contact. The next clinician must know before any drainage.
- The 2026 survey is why we labour this: the gap in deactivation confidence is real. Next: all three spaces on the model, four minutes.

32 UroOps 3D: anatomy and component routes

46:00 – 50:00 · 4:00

- Open the live page. Say it is a teaching schematic with clinical review pending: relationships, not validated anatomy.
- Stop one, the cuff: viable bulbar tissue and its spongiosal support; release traction before measuring.

- Stop two, the dorsal plane: controlled dissection in a normal plane, a change of strategy in a scarred one.
- Stop three, the pump: can he find it and reach its controls seated and standing? Stop four, the balloon: after pelvic surgery the space is not the textbook space.
- Trace the tubing back, recheck connections, cycle, confirm an open cuff and deactivation. If the page fails, use the spoken fallback in the full script.
- Next: the operation is finished, and the first weeks decide whether it was worth doing.

After the implant

50:00 – 60:00 · 10 minutes · slides 33 to 38

Section slide: Hand over the risk, diagnose late leakage before revising, and say where the evidence stops.

33 Aftercare, activation and urgent review

50:00 – 51:00 · 1:00

- After an uncomplicated implant, remove the catheter early and check voiding; retention means assessment and safe drainage, not blind attempts.
- Day-case is feasible in selected pathways: the BAUS audit discharged 88 per cent, and the meta-analysis found no significant excess of retention. Selected, not universal.
- Pain, redness, discharge, fever or voiding difficulty come back to a specialist. Infection or erosion generally means explantation, not antibiotics.
- ContiClassic: four to six weeks deactivated; at activation he demonstrates the device and we verify emptying.
- Device card and clear instructions: alert clinicians before any catheter or scope. Next: our last patient did everything right for three years.

34 Case 4: leakage after three good years

51:00 – 52:30 · 1:30

- Three good years, now leaking again. Would you book a smaller cuff? [Pause five seconds.]
- Separate persistent leakage from true recurrence. Activity, urgency, overflow or difficulty cycling? Watch his technique; check the device state.
- Urine and residual, examine the components, deactivated cystoscopy for erosion, then fluid loss, mechanics and bladder function.
- Bergeson: atrophy in eight per cent of revisions, the balloon in a third; the 2026 lab study shows balloons stiffen and thin with age. A mechanism, not a replacement rule.
- Options run from retraining and bladder treatment to component revision or whole-system replacement. Second implants: 68 per cent reintervention-free at five years.
- Next: so where does all this leave us? What is established, and what honestly is not.

35 Established benefit and unresolved questions

52:30 – 53:15 · 0:45

- AUS delivers substantial benefit, with revision and erosion part of its lifetime burden.
- Tissue preservation, contamination control and planned instrumentation have strong rationale; the individual technical details often lack comparative evidence.
- We cannot name one best salvage for a fragile urethra, one optimal irrigant, or a proven preparation duration.
- New devices, including the first electronic sphincters, need comparable outcome evidence before engineering becomes a clinical claim.
- Make that uncertainty visible in consent and planning. Next: the hour in ten decisions. Five belong before theatre.

36 Five decisions before implantation

53:15 – 54:00 · 0:45

- One: demonstrated stress leakage and an improvement that matters to him.
- Two: watch him use a pump, and judge access over years, not months.
- Three: investigations that answer explicit questions, acknowledging that the guidelines differ.
- Four: keep overactivity, underactivity and poor compliance separate. Five: a stable outlet and a plan for future instrumentation.
- Applied to the cases: optimise case one, investigate case two, stage case three. Next: five decisions protect the result once the implant is open.

37 Five decisions protecting the result

54:00 – 55:00 · 1:00

- Six: antisepsis correct for the product, anatomy and pathway. A fifteen-minute ritual is not a substitute.
- Seven: preserve the tissue you need and measure what the cuff will actually enclose.
- Eight: the instructions for the device being implanted, and explicit stop conditions for injury or contamination.

- Nine: confirm the cuff state, drainage and his ability to use the device; make instrumentation precautions explicit.
- Ten: when leakage returns, diagnose before choosing a smaller cuff or a higher pressure. Next: that is our ten. The last five minutes are yours.

38 Discussion

55:00 – 60:00 · 5:00

- We would like these minutes on decisions, not preferences. Pick a case, say what you would do, and name the finding that would change your plan.
- Take one view on bladder dysfunction or urethral stability, then one contrasting view. Ask: guideline requirement, device instruction, or interpretation?
- Then one question on operative practice, preparation or failure. Keep contributions short enough to allow disagreement.
- A show of hands is opinion, not evidence. Beyond the retrieved evidence, state the uncertainty precisely.
- Close: the useful differences are now clear, what the evidence supports, what the device requires, and where judgement stays decisive.